

JAYANTA BANIK

+1 (760) 672-2203 | jayantabanik.ml@gmail.com | linkedin.com/in/jayantabanik/ | github.com/jayanta-banik | jayantabanik.com/scholar

PROFESSIONAL EXPERIENCE

Aidar Health Inc.

Data scientist

February 2025 - June 2026

Columbia, MD, USA

- Developed Natural Language Processing pipelines for clinical note analysis, processing approximately 11M notes, implementing personally identifiable information (PII) masking and HIPAA compliant data workflows integrated with PCC and MatrixCare (two widely used electronic health record systems), and (International Classification of Diseases) ICD-10 mapping for clinical terminology standardization, ensuring near real-time data synchronization with CMS data sources.
- Reduced cloud spend by 48% (\$9.6K annually) through data analytics-driven capacity planning, usage trend analysis, rightsizing, and reserved compute capacity optimization across cloud workloads.
- Reduced signal processing pipeline latency from 45s to 3s by profiling Python bottlenecks, vectorizing core workflows, and porting performance-critical modules to C++, achieving a 15× runtime improvement.
- Deployed machine learning models, Prophet and LightGBM with MLFlow, forecasting risk in medication interference and vital signals, reducing inference latency by 12%; optimized model serving and cost-per-inference using tag-based allocation and rightsizing, lowering operational expenses by approximately \$250/month.
- Architected an end-to-end low-latency RAG and knowledge graph system, boosting AI-driven LLM responses and delivering real-time action recommendations for nurses, achieving a 75% reduction in care response latency and a 15-hour decrease in turnaround time.
- Designed evaluation methodologies for ML and AI systems including accuracy metrics, human review, annotation, error analysis, and stress testing. Automated testing suite to benchmark pre-deployment as part of CI/CD pipeline.

USDA ARS Salinity Lab

Graduate Student Researcher

September 2022 - December 2024

Riverside, CA, USA

- Designed and implemented a Sensor Directed Sampling algorithm in Python, utilizing statistical experimental design principles to enhance sub-sampling efficiency by 28%, which significantly improved data quality for agent-based methods in deep learning applications.
- Developed an active learning algorithm with a pattern-based architecture (Strategy, Factory) that reduced labeled data requirements for multi-sensor remote sensing models by 65%, facilitating more efficient training for deep learning models.
- Engineered a Spatial Time Series Analysis using ARIMA and SARIMAX, reducing sensor sampling costs on agricultural fields by \$64K/year while achieving a minimum of 94% spatial coverage accuracy, thereby supporting robust experimental design and response surface sampling for field-scale crop monitoring.

SKILLS

Programming: Python, R, SQL, JavaScript, C/C++, CUDA, Arduino, MATLAB, SAS

Data and Infrastructure: Postgres, pgVector, MySQL, Pinecone, NoSQL, Vector Databases, REST APIs, Chroma, Apache Spark, SAS

Data Architecture: Structured and unstructured data pipelines, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP) servers

DevOps: CI/CD, Docker, Git, Operations Research, ReAct pattern, Apache Kafka, Github Actions, SRE, SLA

AI: LLM, Pytorch, Tensorflow, Computer Vision, Scikit-learn, Data Analysis, MLOps, OpenCV, Machine Learning, NLTK, OpenAI, Keras, Neural Networks, Reinforcement Learning, SpaCy, Guardrails, Responsible AI, Model Evaluation and Monitoring, A/B Testing, Transformers, Data Governance

Cloud Infrastructure: SageMaker, Terraform, Lambda, Serverless, EC2, ECS, EKS, AWS, Databricks

Visualization: Matplotlib, Seaborn, ArcGIS Pro, KPI Dashboarding, Power BI, Power Automate

Agentic AI Tools: Spec-Kit, LangChain, LangGraph, Datadog, PostHog, Amazon SageMaker Model Monitor

Data Science: Scikit-learn, Xgboost, CatBoost, LightGBM, JAX, Dask-ML, OpenCV, Hugging Face Transformers, Pandas, NumPy, SciPy, PySpark

Soft Skills: Cross-functional Collaboration, Problem Solving, Ownership, Stakeholder Communication, Analytical Thinking, Project Coordination

EDUCATION

University of California - Riverside

Master's, Computer Engineering

September 2022 - September 2024

West Bengal University of Technology

Bachelor's, Computer Engineering

June 2017 - July 2021

CERTIFICATIONS

- Google TensorFlow Developer License, Google, Issued March 2024, Expires April 2027
- Machine Learning Specialization, [DeepLearning.AI](#) and Stanford Online, February 2024
- Deep Learning Specialization, [DeepLearning.AI](#), February 2024
- Genomic Data Science Specialization, Johns Hopkins University, January 2021
- Deep Learning and Computer Vision: OpenCV, SSD, and GANs, Udemy, February 2020

PROJECTS AND OUTSIDE EXPERIENCE

DNA BERT

April 2026 - Present

- Designed and trained a custom DNA language model using k-mer tokenization, masked-language modeling, distributed training, and supervised fine-tuning on genomic classification tasks, enhancing the model's adaptability in agent-based methods.
- Engineered a large-scale distributed infrastructure using PySpark to facilitate the pretraining of a Large Language Model (LLM) on a DNA/Protein dataset comprising 64 billion datapoints, achieving a remarkable 95% accuracy in antimicrobial resistance classification.

- Architected and deployed a backend pipeline leveraging PyTorch, optimized for fine-tuning a GPT model on DNA editing tasks, incorporating feature engineering of phenotypes to enhance model performance and scalability, aligning with agent-based methods.
- Investigated transformer-based representations of DNA and protein sequences for antimicrobial resistance gene classification, functional annotation, and sequence-level prediction, contributing to advancements in LLM applications.

Isolated Sign Language Recognition

May 2023 - July 2023

- Engineered a robust backend architecture leveraging Python and SQL for seamless integration of video processing and NLP components, driving a 30% improvement in sign language recognition efficiency.
- Designed and deployed a production-ready API that optimizes data flow between image processing and NLP, contributing to the open-source community by sharing the model on Hugging Face.

FAMNet (Agentic AI Framework)

January 2023 - March 2023

- Developed a scalable **Mixture of Experts** (MOE) architecture that achieved LLM performance on the GLUE benchmark while reducing model size to T5-small scale, significantly boosting backend efficiency by 28%.
- Finetuned BERT for dynamic task inference using agent-based methods, enhancing contextual tool selection efficacy through rigorous evaluation and version control practices. Implemented an in-memory vector cache for quick tool selection, leading to a 25% increase in processing speed and improved accuracy for deterministic tasks, while ensuring thorough code review processes.

Emotional AI

September 2022 - December 2022

- Engineered a hybrid deep learning architecture that combines a BERT-based multi-label emotion classifier with a BART-based generator, significantly improving the accuracy of overlapping emotion detection and enabling sophisticated emotional text generation.

Disaster Prediction Using Natural Language Processing

April 2021 - July 2021

- Employed natural language processing techniques to classify tweets and social media posts, successfully identifying credible disaster information and reducing misinformation by filtering out fake or clickbait content.

Image Upscaling Using Generative Models

April 2020 - July 2020

- Leveraged Generative Adversarial Networks (GANs) to significantly improve image quality, removing reliance on interpolation and traditional filtering techniques.

Memory Cycle for Agentic AI

January 2026 - March 2026

- Developed an MCP server for GitHub Copilot, integrating specification-driven agents to automate repository-aware software development workflows, enhancing efficiency by 30%.
- Built a hybrid RAG and knowledge-graph retrieval system using PageIndex-based repository indexing, improving contextual accuracy by 25% and significantly reducing hallucinations in responses.
- Implemented a robust AI security and data-masking layer using LangChain and LangGraph, contributing to responsible AI and data governance by protecting sensitive information and validating agent interactions.
- Added a TurboVec-based validation layer to detect and mitigate prompt poisoning, enhancing the robustness and security of agentic AI workflows by 40%.

PUBLICATIONS

- Near-ground microwave radiometry for on-the-go surface soil moisture sensing in micro-irrigated orchards in California. Agrosystems, Geosciences & Environment, 2025.
- Field deployment of an agricultural robot for informed sampling of soil apparent electrical conductivity using aisle graphs. Precision Agriculture '25, 2025.
- Sensor Directed Sampling. Master's Thesis, University of California, Riverside, 2024.
- Innovative Data Integration for On-the-Go Plant and Soil Sensing: The Short Vectorization Approach. ASA, CSSA, SSSA International Annual Meeting, 2023.
- Study of Dependency on Number of LSTM Units for Character-Based Text Generation Models. IEEE ICCSEA, 2020.

ACTIVITIES AND AWARD

- **International Student Peer Mentor Program**, University of California, Riverside, September 2023: Mentored international students as they adjusted to new academic and cultural environments.
- **Second Place**, Best Poster Presentation, University of California, Riverside, May 2023: Presented "An Innovative Data Integration: The Short Vectorization Approach" at the Precision Agriculture Workshop.
- **Team Leader, 2nd place**, Smart India Hackathon 2020, Government of India, August 2020: Led Team Tesla to a national second-place finish for the Clean Ghats of Varanasi solution.
- Technical Workshop Participant, IIT (B.H.U.), IIT Delhi, IIT Kharagpur, and MyWBUT, 2017 – 2019: Completed workshops in artificial intelligence, internet of things, robotics, autonomous robotics, motion-controlled robotics, voice-controlled robotics, cloud computing, big data, and Hadoop.
- Participant, Google Summer of Code, CERN, April 2018.
- **First Prize**, Inter-College Physics Exhibition, IIT (B.H.U.), June 2015: Awarded at Jigyasa 15 by Dr. H. C. Verma.
- **Gold Medal**, St. Johns School DLW, October 2014: Recognized for excellence in chess and table tennis.

COMMUNITY INVOLVEMENT AND INTERESTS

- **Founder, Joy-Bot Initiative**: Designed and donated an interactive nursing-assistance robot for children and elderly care homes at Anandamayee Ashram, Varanasi, during COVID-19.
- **Vice President, Rotary Club of Varanasi Chapter**: Led educational initiatives mentoring economically disadvantaged students in STEM.
- **International Peer Mentor, University of California, Riverside**: Guided incoming international students through academic and cultural adaptation.
- **Advocate for Ethical and Inclusive AI, MIET Jammu and Kashmir**: Led teaching and community outreach in AI literacy.
- **Guest Instructor, Peer-Learning Sessions, University of California, Riverside**: Conducted workshops on Python for machine learning and deep learning model design.
- **Astrophotography and Astrophysics Enthusiast**: Combining patience and technical curiosity with an appreciation for the cosmos.